

# Mechanism Graphs for the LHC Black-Hole Safety Case

## Static audit report

### Abstract

Scientific papers contain an attribution layer and an equation layer. The first records sources and claims. The second records the mathematical steps that could make a claim physically true. This report applies that separation to the LHC microscopic-black-hole safety debate using a static public audit run. The selected corpus contains 492 arXiv sources, 377 provenance claim nodes and 1,408 equation witnesses. Formula gates reduce the equation layer to 204 usable mechanism nodes and 35 evidence-grade case nodes. The branch split is sharp: 0 source-local formula-clean direct LHC-safety mechanisms, 1 collider-threshold hook and 28 astrophysical black-hole analogues. The useful object is a translation map: accretion, evaporation, capture, mass growth and compact-object survival mechanisms must be carried explicitly into the collider branch before they can support a safety argument.

## 1 Main Claims and Evidence Register

1. **The selected corpus supports a selected-corpus finding.** Evidence: 492 selected sources from the high-recall scan. Interpretation: The scan is broad enough for this public audit object. A universal absence claim would require the full literature and non-arXiv safety reports.
2. **The provenance layer is mostly attribution.** Evidence: 869 provenance nodes and 377 source-to-claim edges. Interpretation: It organizes who said what and where. Equation support has to be tested in a separate graph.
3. **The mechanism layer is much stricter than the provenance layer.** Evidence: 1408 equation witnesses narrowed to 204 usable mechanism nodes. Interpretation: Most extracted text is retained as audit material or rejected by formula-quality gates.
4. **No direct LHC-safety equation branch passed the strict gate.** Evidence: 0 direct LHC-safety mechanism nodes. Interpretation: The absence is defined by the selected corpus and the source-local equation gate.
5. **The populated branch is adjacent black-hole physics.** Evidence: 28 astrophysical analogues and 1 collider-threshold hook. Interpretation: The audit becomes mechanism translation from accretion, evaporation, capture and compact-object survival into the collider branch.
6. **The dominant mechanism routes are operator/closure/transport.** Evidence: constraint closure=165, spectral operator=160, transport flow=95 Interpretation: The graph records equation roles and route co-activation.
7. **Sparse attention supports the branch split.** Evidence: spectral operator=34, constraint closure=33, transport flow=25 Interpretation: The sparse audit is a consistency check over static graph artifacts.

## 2 Data Object

The run directory is `runs/lhc_black_hole_audit_500k_strict`. It contains derived public artifacts: `manifest.json`, `provenance_graph.json`, `equation_witnesses.json`, `operational_graph.json`, `equation_mechanism_graph.json` and `sparse_attention_audit.json`. These files are enough to

## two different graphs in the same papers

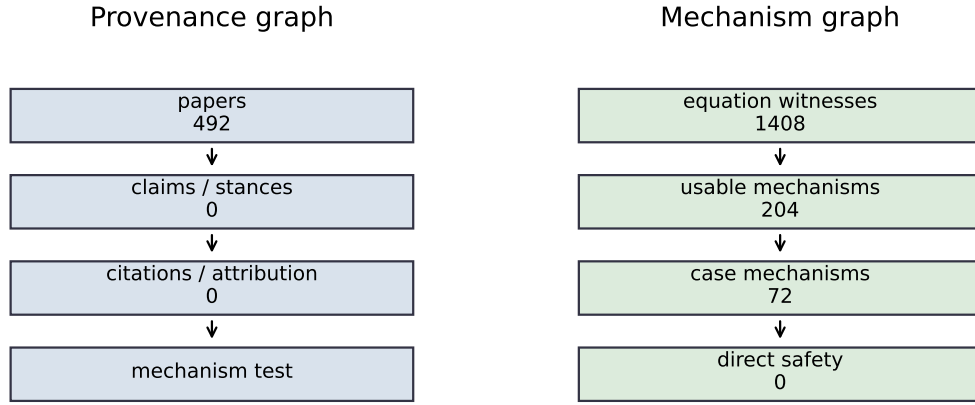


Figure 1: Two graph layers in the same selected archive. Provenance organizes attribution and claims. The mechanism graph records which equation-level branches are present.

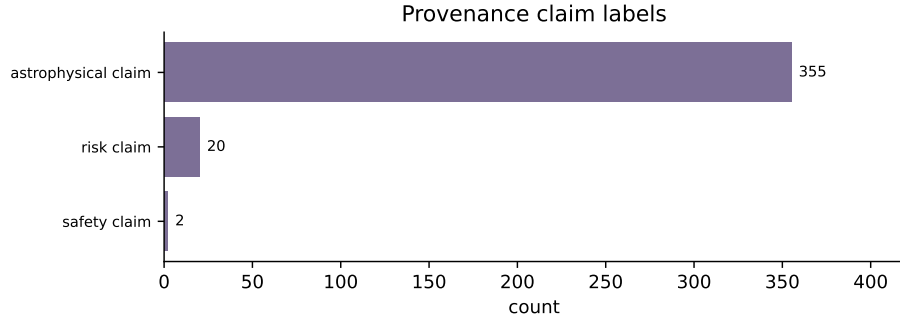


Figure 2: Claim labels in the provenance layer. This layer is an attribution record. Equation support is tested separately.

inspect the report. Private fingerprints, model checkpoints and core pipeline files are outside this repository.

### 3 Provenance Layer

The provenance layer contains 869 nodes and 377 source-to-claim edges. Claim labels in this selected corpus are:

| Quantity            | Count |
|---------------------|-------|
| astrophysical claim | 355   |
| risk claim          | 20    |
| safety claim        | 2     |

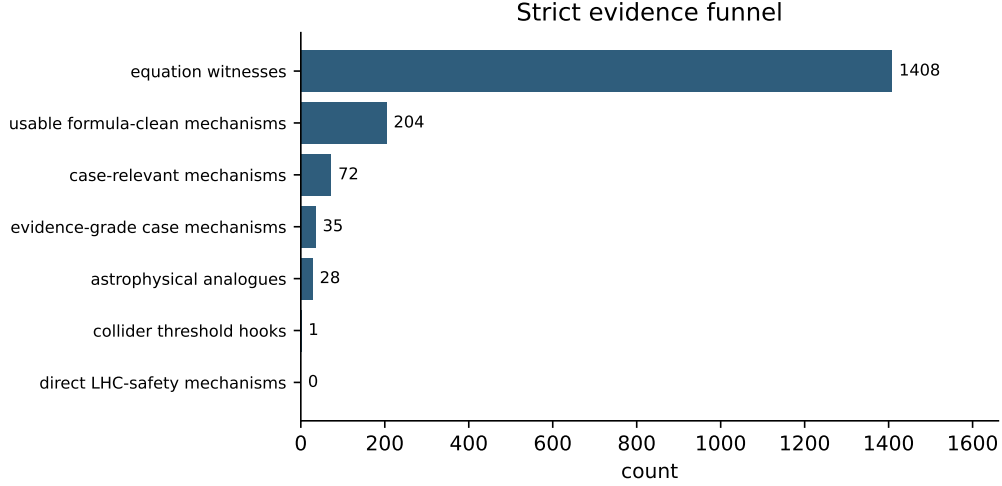


Figure 3: Strict evidence funnel from equation witnesses to case-specific mechanism receipts. The sharp drop is expected: the gate rejects prose fragments, local word matches without a relation operator, and formula windows without case-local mechanism content.

## 4 Mechanism Layer and Gates

The mechanism layer begins with local equation witnesses and applies formula-quality, case-relevance and route gates. The public audit uses six route labels:

|                     |                                                               |
|---------------------|---------------------------------------------------------------|
| transport/flow      | $\partial_t q + \nabla \cdot J = S,$                          |
| constraint/closure  | $C(q, J, \lambda) = 0,$                                       |
| spectral/operator   | $Lq = \lambda q,$                                             |
| boundary/weak form  | $\int_{\Omega} \phi Lq = \int_{\Omega} \phi f, \quad Bq = b,$ |
| commutator/residual | $[A, B] = AB - BA,$                                           |
| discrete protocol   | $x_{n+1} = \Phi(x_n, u_n).$                                   |

These route labels are audit gates. They separate formula-clean mechanism receipts from word matches and bibliographic residue.

## 5 Run Counts

| Quantity                            | Count |
|-------------------------------------|-------|
| selected sources                    | 492   |
| provenance claims                   | 377   |
| source equation witnesses           | 1,408 |
| usable mechanism nodes              | 204   |
| case-relevant mechanism nodes       | 72    |
| evidence-grade case nodes           | 35    |
| direct LHC-safety mechanism nodes   | 0     |
| collider-threshold hooks            | 1     |
| astrophysical analogues             | 28    |
| source-local route-transition edges | 120   |
| rich cross-source route analogues   | 130   |

## 6 Mechanism Distribution

The usable mechanism layer is dominated by closure, spectral/operator and transport routes:

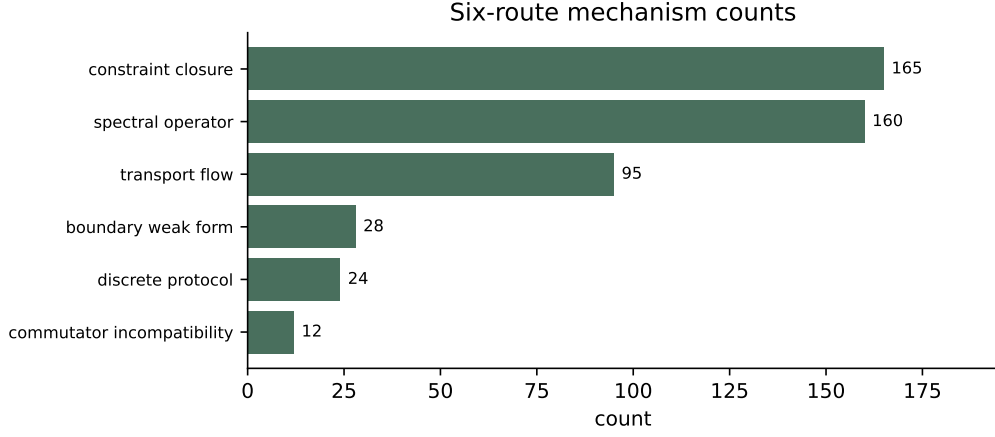


Figure 4: Six-route counts among non-artifact constructor pairs. The reusable layer is a small set of equation operations.

| Quantity                   | Count |
|----------------------------|-------|
| constraint closure         | 165   |
| spectral operator          | 160   |
| transport flow             | 95    |
| boundary weak form         | 28    |
| discrete protocol          | 24    |
| commutator incompatibility | 12    |

Constructor-role and transition-label diagnostics give the same picture:

| Quantity                | Count |
|-------------------------|-------|
| operator apparatus      | 142   |
| selector                | 106   |
| real substrate geometry | 94    |
| closure constraints     | 68    |
| readout current         | 45    |
| protocol order          | 25    |

## 7 LHC Case Branches

Evidence-grade branch counts are:

| Quantity                          | Count |
|-----------------------------------|-------|
| stable growth or capture branch   | 29    |
| astrophysical black hole analogue | 28    |
| evaporation branch                | 12    |
| production threshold branch       | 1     |

The collider branch first requires a production or event-selection condition, for example

$$\sqrt{s} > M_D.$$

If evaporation dominates, a local receipt should support a mass-loss or lifetime ordering such as

$$\frac{dM}{dt} < 0, \quad \tau_{\text{evap}} \ll \tau_{\text{capture}}.$$

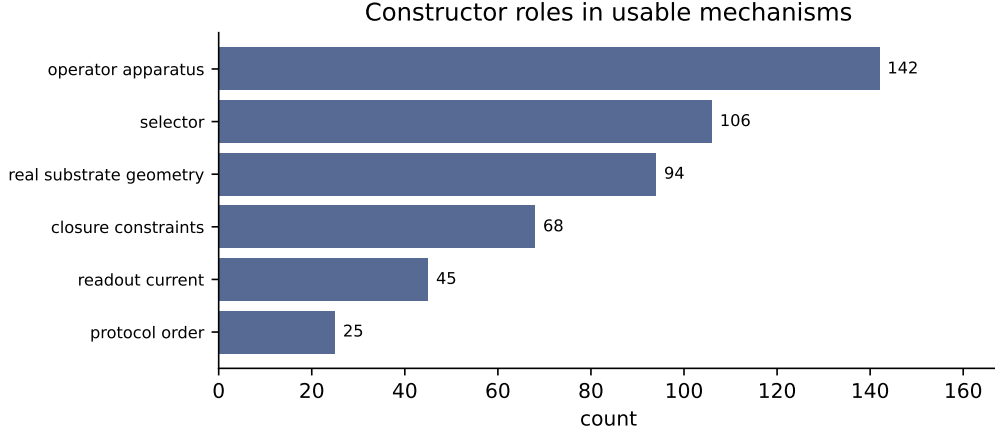


Figure 5: Constructor roles present in usable mechanisms. Operator apparatus, selector and real substrate geometry appear as separable roles.

If a stable object is assumed, the mechanism changes to stopping, capture and growth:

$$\frac{dM}{dt} = \rho \sigma(M) v.$$

Compact-object survival then becomes an external constraint on the same mechanism:

$$t_{\text{grow}} < t_{\text{WD}}$$

would be inconsistent with observed white-dwarf survival if the same production, capture and growth mechanism were realized in nature. The static graph found adjacent receipts for this translation problem. It did not find a direct formula-clean collider-safety branch.

## 8 Sparse-Attention Check

The sparse-attention audit reads the static mechanism graph and measures route co-activation in the evidence-grade case layer.

- Evidence-grade routes: spectral operator=34, constraint closure=33, transport flow=25, boundary weak form=4.
- Astrophysical analogues concentrate on spectral operator=0.35, constraint closure=0.34, transport flow=0.26, boundary weak form=0.04.
- The production-threshold branch is small and separate: boundary weak form=0.50, constraint closure=0.50.
- The strongest route co-activations are constraint closure + spectral operator=32, spectral operator + transport flow=25, constraint closure + transport flow=23.

## 9 Equation Receipts

### 9.1 Direct LHC-Safety Receipts

*No receipt passed the current gate.*

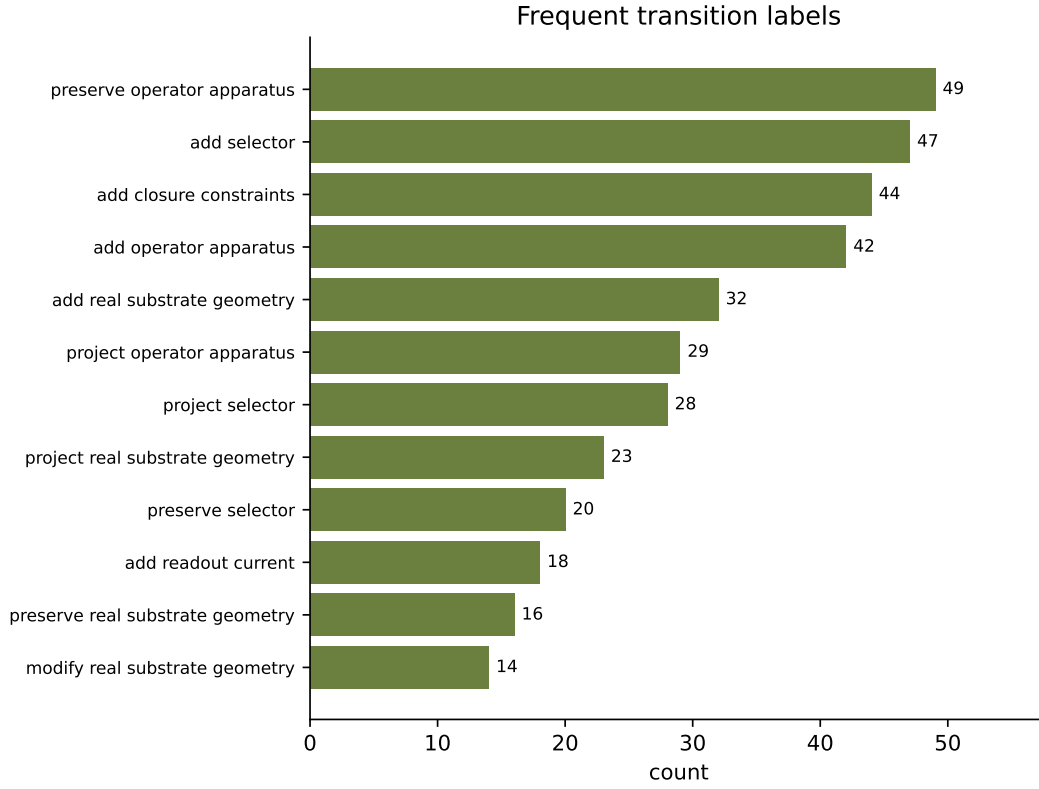


Figure 6: Frequent transition labels. Preservation, addition and projection of operator, selector and closure roles are common moves in the equation layer.

## 9.2 Collider-Threshold or Event-Selection Hook

| Source              | Routes                             | Branch labels                 | Formula witness                          |
|---------------------|------------------------------------|-------------------------------|------------------------------------------|
| 0904.0230<br>E00053 | / constraint<br>boundary weak form | closure,<br>production branch | threshold<br>$ y_{\gamma\gamma}  \leq 1$ |

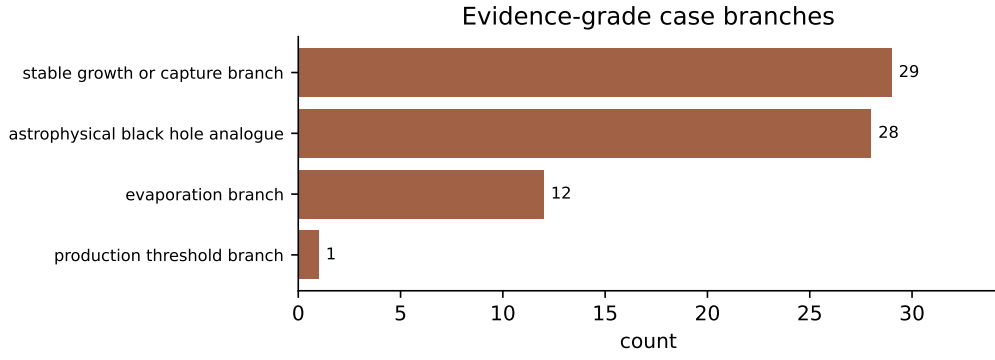


Figure 7: Evidence-grade case branches. The populated branch is adjacent astrophysical black-hole physics; the direct LHC-safety branch remains empty under the strict gate.

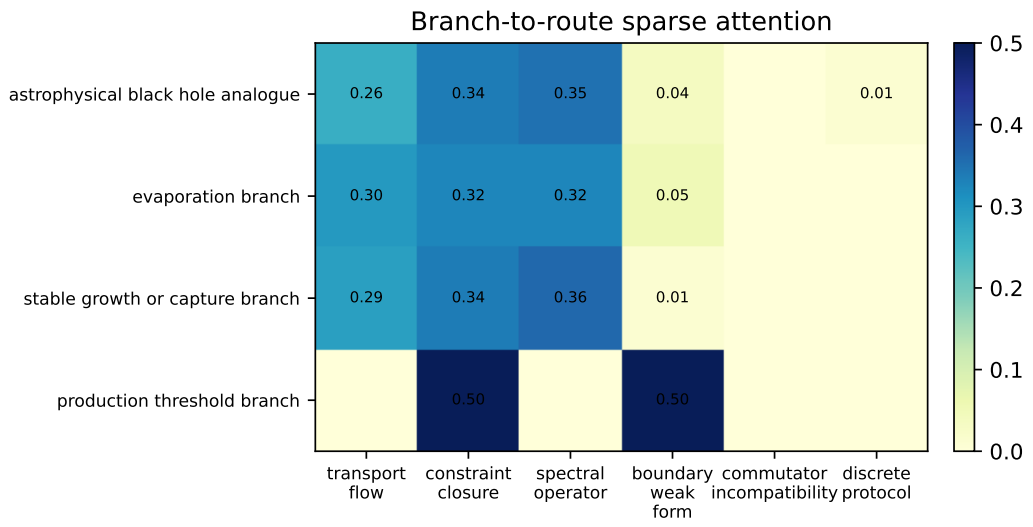


Figure 8: Branch-to-route sparse attention. Astrophysical analogues activate operator, closure and transport routes; the production-threshold branch is separated.

### 9.3 Astrophysical Black-Hole Analogues

| Source                    | Routes                                                                     | Branch labels                                                                          | Formula witness                                                                                                                                                      |
|---------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0807.3458<br>E00034       | / transport flow, constraint closure, spectral operator                    | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $3 \times 10^{-13} \mathrm{M}_{\odot} \mathrm{yr}^{-1}$<br>$\lesssim \angle \dot{M}_{\mathrm{long}}$<br>$\angle \lesssim 1 \times 10^{-10} \mathrm{M}_{\odot} \dots$ |
| 0807.3458<br>E00035       | / transport flow, constraint closure, spectral operator                    | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $4 \times 10^{-14} \mathrm{M}_{\odot} \mathrm{yr}^{-1}$<br>$\lesssim \angle \dot{M}_{\mathrm{long}}$<br>$\angle \lesssim 2 \times 10^{-11} \mathrm{M}_{\odot} \dots$ |
| astro-ph0212297<br>E01142 | / constraint closure, transport flow, spectral operator                    | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $B_c \simeq 10^{16} \mathrm{G} \left( \frac{7M_{\odot}}{M_H} \right) \left( \frac{6M_H}{R} \right)^2 \left( \frac{M_T}{0.03M_H} \right)^{1/2}$                       |
| 1307.7685<br>E00825       | / constraint closure, transport flow, spectral operator                    | astrophysical black hole analogue, stable growth or capture branch                     | $\Delta M_{\mathrm{BH}} < 5 \times 10^{-4} M_{\mathrm{BH}}$                                                                                                          |
| 1207.1244<br>E00797       | / constraint closure, transport flow, spectral operator, discrete protocol | astrophysical black hole analogue                                                      | $M_1 = 0.9_{-0.3}^{+4.6} M_{\odot}$                                                                                                                                  |
| 1604.02455<br>E01020      | / constraint closure, transport flow, spectral operator                    | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $4000 M \sim 60 (M_{\mathrm{rm}} / 1.625 M_{\odot})$                                                                                                                 |
| 1604.02455<br>E01021      | / constraint closure, transport flow, spectral operator                    | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $\Delta t \sim 0.1 (M_{\mathrm{rm}} / 1.625 M_{\odot})$                                                                                                              |
| astro-ph0212297<br>E01141 | / constraint closure, spectral operator, transport flow                    | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $\frac{\mathcal{E}_B}{\mathcal{E}_k} < 0.1$                                                                                                                          |



## 9.4 Stable Growth or Capture Branch

| Source                    | Routes                                                  | Branch labels                                                                          | Formula witness                                                                                                                                                 |
|---------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0807.3458<br>E00034       | / transport flow, constraint closure, spectral operator | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $3 \times 10^{-13} \mathrm{M} \dot{\mathrm{yr}}^{-1} \lesssim \angle \dot{M} \mathrm{long} \angle \lesssim 1 \times 10^{-10} \mathrm{M} \dot{\mathrm{yr}}^{-1}$ |
| 0807.3458<br>E00035       | / transport flow, constraint closure, spectral operator | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $4 \times 10^{-14} \mathrm{M} \dot{\mathrm{yr}}^{-1} \lesssim \angle \dot{M} \mathrm{long} \angle \lesssim 2 \times 10^{-11} \mathrm{M} \dot{\mathrm{yr}}^{-1}$ |
| astro-ph0212297<br>E01142 | / constraint closure, transport flow, spectral operator | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $B.c \lesimeq 10^{16} \mathrm{G} \left( \frac{7M_{\odot}}{M_H} \right) \left( \frac{6M_H}{R} \right)^2 \left( \frac{M_T}{0.03M_H} \right)^{1/2}$                |
| astro-ph0109539<br>E01098 | / constraint closure, spectral operator, transport flow | stable growth or capture branch                                                        | $\angle M \angle \sim 9 \mathrm{M} \dot{\mathrm{yr}}^{-1}$                                                                                                      |
| 1307.7685<br>E00825       | / constraint closure, transport flow, spectral operator | astrophysical black hole analogue, stable growth or capture branch                     | $\Delta M_{\mathrm{BH}} < 5 \times 10^{-4} M_{\mathrm{BH}}$                                                                                                     |
| 1604.02455<br>E01020      | / constraint closure, transport flow, spectral operator | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $4000M \sim 60(M_{\mathrm{rm}} \mathrm{NS} / 1.625M_{\odot})$                                                                                                   |

## 9.5 Evaporation Branch

| Source                    | Routes                                                  | Branch labels                                                                          | Formula witness                                                                                                                                                                         |
|---------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0807.3458<br>E00034       | / transport flow, constraint closure, spectral operator | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $3 \times 10^{-13} \mathrm{M} \dot{\mathrm{yr}}^{-1} \lesssim \frac{\dot{M}}{\mathrm{long}} \frac{\mathrm{angle}}{\mathrm{lessim} 1} \times 10^{-10} \mathrm{M} \dot{\mathrm{yr}}^{-1}$ |
| 0807.3458<br>E00035       | / transport flow, constraint closure, spectral operator | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $4 \times 10^{-14} \mathrm{M} \dot{\mathrm{yr}}^{-1} \lesssim \frac{\dot{M}}{\mathrm{long}} \frac{\mathrm{angle}}{\mathrm{lessim} 2} \times 10^{-11} \mathrm{M} \dot{\mathrm{yr}}^{-1}$ |
| astro-ph0212297<br>E01142 | / constraint closure, transport flow, spectral operator | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $B.c \lesimeq 10^{16} \mathrm{G} \left( \frac{7M_{\mathrm{odot}}}{M_H} \right) \left( \frac{6M_H}{R} \right)^2 \left( \frac{M_T}{0.03M_H} \right)^{1/2}$                                |
| 1604.02455<br>E01020      | / constraint closure, transport flow, spectral operator | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $4000M \sim 60(M_{\mathrm{rm}} \mathrm{NS}/1.625M_{\mathrm{odot}})$                                                                                                                     |
| 1604.02455<br>E01021      | / constraint closure, transport flow, spectral operator | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $\Delta t \sim 0.1 (M_{\mathrm{rm}} \mathrm{NS}/1.625M_{\mathrm{odot}})$                                                                                                                |
| astro-ph0212297<br>E01141 | / constraint closure, spectral operator, transport flow | astrophysical black hole analogue, evaporation branch, stable growth or capture branch | $\frac{\mathcal{E}_B}{\mathcal{E}_k} < 0.1$                                                                                                                                             |

## 10 Interpretation

Provenance remains necessary for attribution, incentives and chronology. It cannot settle a physics mechanism by itself. In the selected LHC black-hole corpus, claim extraction finds many statements and few safety or risk claims. Equation gating finds a different object: an empty direct collider-safety branch under the source-local gate, plus adjacent black-hole mechanisms that can constrain the collider branch if the transfer assumptions are stated.

## 11 Scope

All claims in this report are scoped to the selected corpus and the public gates. A complete safety review would require a full literature selection, non-arXiv safety reports and human review of the equation receipts. The repository provides a static separation between provenance and mechanism graphs.